Annihilator Sketches and the Complexity of Distributed Vertex Connectivity

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Abstract

We introduce *annihilator sketching*, a technique for handling vertex deletions directly on graph sketches. After the sketches have been formed, a referee can algebraically eliminate all edges incident to a specified set of vertices, without recovering those edges or accessing the original graph. Supporting k vertex deletions incurs only an additive $O(k)$ overhead in sketch dimension.

We apply this technique to vertex connectivity in the distributed graph sketching model, where each vertex of an n-vertex graph simultaneously sends a single message, computed from its neighborhood, to a referee. Annihilator sketching yields protocols for k-vertex connectivity using $O(k \log^3 n)$ bits per vertex with public randomness and $O(\sqrt{kn} \log^{3/2} n)$ bits per vertex with private randomness. Both protocols succeed with high probability and construct sparse subgraphs preserving connectivity after every set of fewer than k vertex failures. The public-randomness bound improves the $O(k \log^4 n)$ bound obtained from Assadi and Shah (SOSA 2023), resolving a question of Assadi. The private-randomness protocol extends the approach of Holm et al. (FOCS 2019) from ordinary connectivity to vertex connectivity. Our public-randomness construction also yields a one-pass dynamic streaming algorithm using $O(kn \log^3 n)$ bits of space.

We complement these randomized protocols with a lower bound of $\Omega(k + \sqrt{n})$ on the maximum message length of any deterministic protocol for k-vertex connectivity, for $1 \leq k \leq n/6$. This bound holds even for deciding whether the graph is k-vertex-connected, without requiring a certificate. In particular, for ordinary connectivity it improves the $\Omega(n^{1/3})$ bound of Robinson and Tan (2026) to $\Omega(\sqrt{n})$.

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1 Introduction

- Testing k-node connectivity.
  - Public Random: $O(k \log^3 n)$
  - Private Random: $O(\sqrt{kn} \log^? n)$
  - Deterministic: $\Omega(k + \sqrt{n})$
- Distinguish k-node connected instances from disconnected instances.
  - Public Random: $O(\log n \log^2(n/k)$
  - Private Random: $O(\min(\sqrt{n}, n/k))$
  - Deterministic: $\Omega(\min(\sqrt{n}, n/k)$

2 Technical Overview

2.1 Upper Bound

2.2 Lower Bound

3 Preliminaries

signed incidence and adjacency matrices

Graphs. Throughout this work, a *graph* is undirected, simple, and unweighted, unless qualified otherwise. All computational problems we consider have a single input graph $G = (V, E)$, where $|V| = n$; we usually identify V with the set $[n] := \{1, \dots, n\}$. Formally, each edge $e \in E$ is a set consisting of exactly two nodes. For a set $Z \subseteq V$, we define $E(Z) = \{e \in E : e \subseteq Z\}$ and $\partial Z = \{e \in E : |e \cap Z| = 1\}$, i.e., the set of edges lying inside Z and those "leaving" Z, respectively. We write $G[Z]$ for the subgraph of G induced by Z and $G \setminus Z$ for the subgraph obtained by deleting Z; note that $G \setminus Z = G[V \setminus Z]$. For a set $F \subseteq E$, we write $G - F$ for the (non-induced) subgraph $(V, E \setminus F)$.

The Distributed Graph Sketching Model. We study problems where the desired output is some possibly multi-valued function ϕ of the input graph G.[1] We focus on *distributed* graph algorithms for such problems.

Our basic model of computation is a communication game among $n + 1$ players named $\textsc{plr}_1, \dots, \textsc{plr}_n$ and $\textsc{coord}$. For each $i \in [n]$, $\textsc{plr}_i$ is a *node player* who receives as input the neighborhood of node $i \in V$, i.e., the set $\mathrm{N}(i) := \{j \in V : \{i, j\} \in E\}$. Note that the knowledge of each edge $e \in E$ is shared between the two node players corresponding to e's endpoints. The special player $\textsc{coord}$ is the *coordinator*, who receives no input, but must eventually output $\phi(G)$. The distributed algorithm for computing $\phi(G)$ is modeled as a *protocol* that operates as follows. There is a public random string R accessible to all players. Each node player $\textsc{plr}_i$ computes a message $M_i = \textsc{msg}_i(\mathrm{N}(i), R) \in \{0,1\}^*$; the messages $M_1, \dots, M_n$ are simultaneously communicated to $\textsc{coord}$. The coordinator then computes a function $\mathrm{out}(M_1, \dots, M_n, R)$ that specifies the output. The functions $\{\textsc{msg}_i\}_{i \in [n]}$ and out completely specify the protocol. Note that the communication in the protocol is *non-interactive*.

Let Π be a protocol for some graph problem ϕ and let $\textsc{msg}_i$, out, and M_i be as above. The error probability $\mathrm{err}(\Pi)$ is defined to be $\max\{\mathrm{err}(\Pi, G) : G \text{ an } n\text{-vertex graph}\}$, where $\mathrm{err}(\Pi, G) = \mathbf{Pr}[\mathrm{out}(M_1, \dots, M_n, R) \neq$

[1] We have chosen to avoid the full formalism of *relations* for capturing problems with multiple correct answers.

$\phi(G)]$. The communication cost is defined to be $\|\Pi\| = \max_{G,R} \sum_{i=1}^{n} \text{len}(M_i)$, the worst-case total length of all messages sent by the node players. Our primary goal in designing efficient protocols is to achieve $\text{err}(\Pi) \leq \delta$—where δ is either a small constant or, better yet, $\delta = n^{-O(1)}$—while also achieving communication cost $\|\Pi\| = o(n^2)$, which is sublinear in the input size.[2]

Problems of Interest. The problems studied in this work are (i) the Boolean GRAPH-CONNECTEDNESS problem (henceforth, IS-CONN), where $\phi(G)$ is TRUE if G is connected and FALSE otherwise; (ii) the CONNECTED-COMPONENTS problem (henceforth, CONN-COMPS), where $\phi(G)$ is the partition of V into the connected components of G, and (iii) the SPANNING-FOREST problem (henceforth, SP-FOREST), where $\phi(G)$ is multi-valued and can be any spanning forest of G. It is immediate that this list of problems has been given in increasing order of difficulty.

4 A New Sketching Primitive

Previous protocols for k-vertex connectivity in [GMT15; AS23], as well as for related problems such as the even more fundamental connectivity problem, [AGM12; HKTZZ19] all rely on maintaining linear sketches of each node's row in the signed incidence matrix of G. The streaming and public randomness dynamic sketching protocols maintain an ℓ_0 sampler sketch of this vector, while the private randomness distributed sketching and deterministic protocols maintain a sparse recovery sketch.

All of these protocols proceed by building up "blobs" of nodes, which eventually form the components of a graph. When using rows of the signed incidence matrix, the useful property is that for an edge $\{u, v\} \in E$ with $u < v$, coordinate $\{u, v\}$ of u's row will have a $+1$ while the same coordinate in v's row will have a -1. Therefore, taking *linear* sketches of these rows, and summing up the sketches from all nodes within a certain blob *cancel out* all of the interior edges, leaving only the boundary edges. Then, ℓ_0 sampling or sparse recovery can be used to recover edges from this boundary, allowing the blob to be built up further.

Our main contribution in this work is to introduce a new primitive based on sketching the signed *adjacency* matrix instead of the signed incidence matrix. Initially, this might seem counterintuitive based on how the protocols operate, since these sketches cannot be summed up to remove interior edges; an edge $\{i, j\}$ will now appear in the jth coordinate of i's row and the ith coordinate of j's row. However, we will show that these sketches can be postprocessed—without knowledge of the graph—such that they act as a sketch of a scaling of the signed incidence matrix. Thus, summing up the sketches of a set of nodes eliminate the set's interior edges. This allows our new primitive to recover the results of [AGM12; HKTZZ19]. Furthermore, the main benefit of this new primitive is that for any set of nodes $F \subseteq V$ with $|F| \leq k$, revealed *after* sketching, the sketches can be processed—again without knowledge of the graph—such that they sketch a scaling of the signed incidence matrix of $G[V \setminus F]$, with only an additive k factor to the sketch dimension. This allows us—as shown in Section 5.1—to solve the k-vertex-connectivity problem *without* any graph subsampling, improving on [AS23] in the dynamic streaming and public randomness distributed sketching settings, and give the first protocol in the private randomness distributed sketching setting.

We now describe how to use our sketching primitive to achieve both sparse recovery and fingerprint sketches. Using these two together also gives us have sparseness detection *and* recovery, which has many uses, such as constructing ℓ_0 samplers. The rest of the section explains how the sketches are computed.

Theorem 1. *Consider a graph G and parameters $c, d, k \in \mathbb{N}$. Knowing only their neighborhood $\text{N}(v)$, each node $v \in V$ can maintain linear sketches $s(v)$ and $f(v)$ using $O((d + k)\log n)$ and $O((c + k)\log n)$ bits respectively, with the following guarantee. For any subset of nodes $F \subseteq V$ with $|F| \leq k$, a sketch $\tilde{s}_F(v)$ can be*

[2] A communication cost of $O(n^2)$ is trivial for any problem ϕ: this large a cost allows each PLR_i to simply send $\text{N}(i)$ as is.

computed from only $s(v)$ and F such that $\tilde{s}_F(v)$ is a d-sparse recovery sketch, allowing the decoding of v's row in the signed incidence matrix of $G[V \setminus F]$. Similarly, $\tilde{f}_F(v)$ can be computed from only $f(v)$ and F such that $\tilde{f}_F(v)$ is a fingerprint sketch of v's row of the signed incidence matrix of $G[V \setminus F]$, with fingerprint collision probability n^{-c}.

Knowing their neighborhood $\mathrm{N}(v)$, each node v can construct their row of the signed adjacency matrix of G, $A^{\pm} \in \{-1, 0, 1\}^{n \times n}$. Call this row $\boldsymbol{a}_v^{\pm}$. We first explain the sparse recovery sketch. Each node maintains the sketch $s(v) = \Psi \boldsymbol{a}_v^{\pm}$, using the sketching matrix $\Psi \in \mathbb{F}_Q^{(2d+k+1)\times n}$ for an appropriately large finite field $\mathbb{F}_Q$. Throughout this paper, we shall take Q to be a prime with $Q = \Theta(n^3)$. Note that this makes $s(v)$ a $(2d + k + 1)$-dimensional sketch vector where each element can be denoted with $O(\log n)$ bits. Formally stated,

$$\Psi = \begin{bmatrix} (a_1)^0 & (a_2)^0 & \cdots & (a_n)^0 \\ (a_1)^1 & (a_2)^1 & \cdots & (a_n)^1 \\ \vdots & \vdots & \ddots & \vdots \\ (a_1)^{2d+k} & (a_2)^{2d+k} & \cdots & (a_n)^{2d+k} \end{bmatrix},$$

where a_i is the ith prime. We use the ith prime for convenience, but note that any set of elements n field elements that satisfy the conditions $a_i \neq a_j$ when $i \neq j$ and $a_i a_j \neq a_k a_l$ when $\{i, j\} \neq \{k, l\}$ would work as well. In contrast, the choice $a_i = i$, which is the usual choice for sparse recovery sketches, would not satisfy the second requirement.

Remark. The annihilator sketch construction does not inherently require Ψ to be a Vandermonde matrix with prime field elements. In fact, we can more generally state the following. For any set $F \subseteq V$ with $|F| \leq k$, denote the submatrix of Ψ on only the columns of F as Ψ_F. Let $\boldsymbol{p}_1, \ldots, \boldsymbol{p}_m \in \ker(\Psi_F^{\mathsf{T}})$ be $m \geq 2d$ linearly independent vectors chosen with knowledge only of Ψ_F, and take $P = [\boldsymbol{p}_1 \cdots \boldsymbol{p}_m]$. Then, the necessary property for Ψ is that for any $2d$ distinct pairs $\{u, v\}$ with $u, v \notin F$, the elementwise product of $(P^{\mathsf{T}}\Psi)$'s u^{th} and v^{th} columns are linearly independent.

For the rest of this section, we shall use for clarity of exposition the explicit Vandermonde construction for Ψ above. This allows us to use an explicit annihilator polynomial in place of the kernel and linearly independent vector construction, as well as leading to efficient and well-known decoding procedures for sparse recovery Efe: citation of a sparse recovery decoder using Vandermonde. Section 5.3 shows a construction for the fingerprint sketch that uses a fully random matrix, which highlights these more general construction ideas instead of the specific structure and properties of the Vandermonde. !!!

In effect, our sketch maintains

$$s(v)_t = \sum_{u \in \mathrm{N}(v)} A^{\pm}_{v,u} (a_u)^t = \sum_{u \in \mathrm{N}(v)} \operatorname{sgn}(u - v)(a_u)^t$$

for each coordinate $t \in \{0, \ldots, 2d + k\}$, where $\operatorname{sgn}(\cdot)$ is the signum function. We zero-index the sketch vector for convenience. Importantly, as a linear sketch, $s(v)$ can be maintained for turnstile streams as well as computed in distributed models.

Now, after the sketch is computed, take a subset of nodes $F \subseteq V$ with $|F| \leq k$. Using the set F, we define the polynomial over $\mathbb{F}_Q$

$$P_F(x) = \prod_{z \in F} (x - a_z) = \sum_{i=0}^{|F|} b_i x^i, \tag{1}$$

with the set of coefficients $\{b_i\}_{i=0}^{|F|}$ obtained by expanding the product. It can be seen that $P_F(a_v) = 0$ if $v \in F$. We now define the sketch $\tilde{s}_F(v)$ such that

$$\tilde{s}_F(v)_t = (a_v)^t P_F(a_v) \sum_{i=0}^{|F|} b_i s(v)_{t+i} \tag{2}$$

for each coordinate $t \in \{0, \dots, 2d+k-|F|\}$. Notice that $\tilde{s}(v)$'s dimension is $|F|$ less than $s(v)$, and that it can be computed using only P_F and $s(v)$, without any knowledge of the graph. Therefore, it can be computed by the coordinator in the distributed sketching model, and at the end of the stream in the turnstile streaming model. We make the following observation about this sketch.

Lemma 4.1. *For any node $v \notin F$ and each coordinate $t \in \{0, \dots, 2d + k - |F|\}$,*

$$\tilde{s}_F(v)_t = \sum_{u \in N(v) \setminus F} \operatorname{sgn}(v-u)(a_v a_u)^t P_F(a_v) P_F(a_u).$$

Proof. We expand the definition from eq. (2) as

$$\begin{aligned}
(a_v)^t P_F(a_v) \sum_{i=0}^{|F|} b_i s(v)_{t+i} &= (a_v)^t P_F(a_v) \sum_{i=0}^{|F|} b_i \sum_{u \in \mathrm{N}(v)} \operatorname{sgn}(v-u)(a_u)^{t+i} \\
&= \sum_{u \in \mathrm{N}(v)} \operatorname{sgn}(v-u)(a_v a_u)^t P_F(a_v) \sum_{i=0}^{|F|} b_i (a_u)^i \\
&= \sum_{u \in \mathrm{N}(v)} \operatorname{sgn}(v-u)(a_v a_u)^t P_F(a_v) P_F(a_u) \\
&= \sum_{u \in \mathrm{N}(v) \setminus F} \operatorname{sgn}(v-u)(a_v a_u)^t P_F(a_v) P_F(a_u),
\end{aligned}$$

where the last equality is because $P_F(a_u) = 0$ when $u \in F$. □

Now, take the signed incidence matrix $B_F^{\pm}$ of $G[V \setminus F]$, and scale each column $\{v, u\}$ by $P_F(a_v)P_F(a_u)$. Note that the scaling itself does not depend on the graph. Then, $\tilde{s}_F(v)$ is the sketch of v's row of this matrix using the sketching matrix

$$\begin{bmatrix}
(a_1a_2)^0 & (a_1a_3)^0 & \cdots & (a_na_{n-1})^0 \\
(a_1a_2)^1 & (a_1a_3)^1 & \cdots & (a_na_{n-1})^1 \\
\vdots & \vdots & \ddots & \vdots \\
(a_1a_2)^{2d+k-|F|} & (a_1a_3)^{2d+k-|F|} & \cdots & (a_na_{n-1})^{2d+k-|F|}
\end{bmatrix},$$

since for a node v, $(B_F^{\pm})_{v,\{v,u\}}$ is 1 if $v < u$ and -1 if $v > u$. Using the facts that $|F| \le k$ and that each product $a_i a_j$ is distinct for pairs $\{i, j\}$, is a d-sparse recovery sketching matrix. Since the scaling itself is nonzero, recovering the underlying vector in the columnwise scaling of $B_F^{\pm}$ allows us to recover the underlying vector in $B_F^{\pm}$ itself. This proves the sparse recovery component of Theorem 1.

Furthermore, since we have a columnwise scaling of the signed incidence matrix, for a subset of nodes $C \setminus V$, summing up $\sum_{v \in C} \tilde{s}_F(v)$ still cancels out the interior edges, leaving only a d-sparse recovery sketch of the boundary edges. Formally, we can state this as follows.

Lemma 4.2. *For any set of vertices $C \subseteq V \setminus F$ and each coordinate $t \in \{0, \dots, 2d + k - |F|\}$,*

$$\sum_{v \in C} \tilde{s}_F(v)_t = \sum_{\substack{\{v,u\} \in E: \\ v \in C, u \notin C \cup F}} \operatorname{sgn}(v-u)(a_v a_u)^t P_F(a_v) P_F(a_u).$$

Proof. We can expand the left hand side as

$$\sum_{v \in C} \sum_{u \in \mathrm{N}(v) \setminus F} \operatorname{sgn}(v-u)(a_v a_u)^t P_F(a_v) P_F(a_u) = \sum_{\substack{(v,u): \{v,u\} \in E, \\ v,u \in C, v < u}} \operatorname{sgn}(v-u)(a_v a_u)^t P_F(a_v) P_F(a_u)$$

$$+ \sum_{\substack{(v,u):\{v,u\}\in E,\\ v,u\in C, v>u}} \operatorname{sgn}(v-u)(a_v a_u)^t P_F(a_v)P_F(a_u)$$
$$+ \sum_{\substack{(v,u):\{v,u\}\in E,\\ v\in C, u\notin C\cup F}} \operatorname{sgn}(v-u)(a_v a_u)^t P_F(a_v)P_F(a_u).$$

The first two sums are over the exact same edges $\{v,u\} \in E$, once from each endpoint. Therefore, the signum function makes sure the first two sums exactly cancel out, leaving only the third. The ordering of the pair (v,u) can also be removed, since each edge is only counted from the endpoint in C. □

Now, we explain the fingerprint sketch. The main new ingredient of this sketch over sparse recovery is its use of public randomness, in order to create sketch vectors with low collision probability. Specifically, each node, using public randomness, will compute a random vector $\boldsymbol{r} \in \mathbb{F}_Q^{\binom{n}{2}}$, where each entry $\boldsymbol{r}_{\{u,v\}}$ is and independently and uniformly drawn field element of $\mathbb{F}_Q \setminus \{0\}$. Then, each node maintains the sketch $f(v) = \Phi \boldsymbol{a}_v^{\pm}$ for a sketching matrix $\Phi \in \mathbb{F}_Q^{(c+k+1)\times n}$. It is intuitive to see Φ as the matrix product of the Vandermonde matrix of primes we used in the sparse recovery sketch and a random diagonal matrix, as

$$\Phi = \begin{bmatrix} (a_1)^0 & (a_2)^0 & \cdots & (a_n)^0 \\ (a_1)^1 & (a_2)^1 & \cdots & (a_n)^1 \\ \vdots & \vdots & \ddots & \vdots \\ (a_1)^{c+k} & (a_2)^{c+k} & \cdots & (a_n)^{c+k} \end{bmatrix} \begin{bmatrix} \boldsymbol{r}_{\{v,1\}} & 0 & \cdots & 0 \\ 0 & \boldsymbol{r}_{\{v,2\}} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \boldsymbol{r}_{\{v,n\}} \end{bmatrix}.$$

Formally, $\boldsymbol{r}_{\{v,v\}}$ is undefined, so we can assume that entry in the diagonal matrix is set to 0. As our graph is simple, it will be multiplied by 0 in the signed adjacency row $\boldsymbol{a}_v^{\pm}$ anyways. This sketch maintains

$$f(v)_t = \sum_{u\in N(v)} \operatorname{sgn}(v-u)\boldsymbol{r}_{\{v,u\}}(a_u)^t$$

for each coordinate $t \in \{0,\dots,c+k\}$.

Now, given a subset $F \setminus V$ with $|F| \le k$, the process of the coordinator mirrors the above case for sparse recovery sketching. We again construct the polynomial $P_F(x)$ exactly as in eq. (1), and use it to compute the sketch $\tilde{f}_F(v)$ such that

$$\tilde{f}_F(v)_t = (a_v)^t P_F(a_v) \sum_{i=0}^{|F|} b_i f(v)_{t+i}$$

for each coordinate $t \in \{0,\dots,c+k-|F|\}$. The equivalent observation to Lemma 4.1 is then the following. The proof proceeds identically, so we do not reproduce it for conciseness.

Lemma 4.3. *For any node $v \notin F$ and each coordinate $t \in \{0,\dots,c+k-|F|\}$,*

$$\tilde{f}_F(v) = \sum_{u\in N(v)\setminus F} \operatorname{sgn}(v-u)\boldsymbol{r}_{\{v,u\}}(a_v a_u)^t P_F(a_v)P_F(a_u).$$

Before explaining the collision probability of this fingerprint sketch, we can also apply the same logic as in Lemma 4.2—using also the fact that public randomness supplies v and u with the same $\boldsymbol{r}_{\{v,u\}}$—in order to see that for any set of vertices $C \subseteq V \setminus F$ and each coordinate $t \in \{0,\dots,c+k-|F|\}$,

$$\sum_{v\in C} \tilde{f}_F(v)_t = \sum_{\substack{\{v,u\}\in E:\\ v\in C, u\notin C\cup F}} \operatorname{sgn}(v-u)\boldsymbol{r}_{\{v,u\}}(a_v a_u)^t P_F(a_v)P_F(a_u).$$

All that remains is to show, for a fixed F of size $|F| \le k$, that the collision probability of $\sum_{v \in C} \tilde{f}_F(v)$, which is the sketch of the actual underlying boundary edge vector of C, and the sketch of a purported underlying boundary edge vector of C is n^{-c}. In particular, by taking C as a singleton, we retrieve the fingerprint component of Theorem 1. For clarity, call the actual boundary edge vector $\boldsymbol{b}$ and the purported boundary edge vector $\boldsymbol{b}'$, such that $\boldsymbol{b} \neq \boldsymbol{b}'$. Call the sketch of $\boldsymbol{b}'$ $\overline{f}$. Notice that this is a specific instantiation of the fingerprint collision probability since the coordinator can detect when a recovered underlying vector does not correspond to a possible boundary edge vector of C, e.g., by containing edges internal or external to C, without needing to consult the fingerprint. In fact, we can think of $\boldsymbol{b}'$ as the actual boundary edge sketch of C under a different purported edge set E' that differs from E in at least the boundary of C.

Proof of collision probability. $\boldsymbol{b} \neq \boldsymbol{b}'$ implies, as explained above, that

$$\left\{\{v,u\} \mid \{v,u\} \in E, v \in C, u \notin C \cup F\right\} \neq \left\{\{v,u\} \mid \{v,u\} \in E', v \in C, u \notin C \cup F\right\}.$$

So, we have that

$$\begin{aligned}
\sum_{v \in C} \tilde{f}_F(v) - \overline{f}_t &= \sum_{\substack{\{v,u\} \in E \setminus E': \\ v \in C, u \notin C \cup F}} \operatorname{sgn}(v-u) \boldsymbol{r}_{\{v,u\}} (a_v a_u)^t P_F(a_v) P_F(a_u) - \sum_{\substack{\{v,u\} \in E' \setminus E: \\ v \in C, u \notin C \cup F}} \operatorname{sgn}(v-u) \boldsymbol{r}_{\{v,u\}} (a_v a_u)^t P_F(a_v) P_F(a_u) \\
&= \sum_{\substack{\{v,u\} \in E \triangle E': \\ v \in C, u \notin C \cup F}} (-1)^{\mathbb{1}\left[\{v,u\} \in E'\right]} \operatorname{sgn}(v-u) \boldsymbol{r}_{\{v,u\}} (a_v a_u)^t P_F(a_v) P_F(a_u) \\
&= \sum_{\substack{\{v,u\} \in E \triangle E': \\ v \in C, u \notin C \cup F}} \left((-1)^{\mathbb{1}\left[\{v,u\} \in E'\right]} \operatorname{sgn}(v-u) (a_v a_u)^t P_F(a_v) P_F(a_u) \right) \cdot \boldsymbol{r}_{\{v,u\}},
\end{aligned}$$

where $\mathbb{1}[\cdot]$ is the indicator function. For convenience, we shall denote this symmetric difference set as

$$\Delta \coloneqq \left\{\{v,u\} \mid \{v,u\} \in E \triangle E', v \in C, u \notin C \cup F\right\}.$$

Now, we can rewrite the event of a collision, meaning $\sum_{v \in C} \tilde{f}_F(v) - \overline{f} = \mathbf{0}$, as $M \boldsymbol{r}_\Delta = \mathbf{0}$ where $\boldsymbol{r}_\triangle \in \mathbb{F}_Q^{|\Delta|}$ is the subvector of $\boldsymbol{r}$ for edges $\{v,u\} \in \Delta$. In turn, $M \in \mathbb{F}_Q^{(c+k-|F|+1) \times |\Delta|}$ is the matrix with

$$M_{t,\{v,u\}} = (-1)^{\mathbb{1}\left[\{v,u\} \in E'\right]} \operatorname{sgn}(v-u) (a_v a_u)^t P_F(a_v) P_F(a_u).$$

Since for each edge $\{v,u\}$, $(-1)^{\mathbb{1}\left[\{v,u\} \in E'\right]} \operatorname{sgn}(v-u) P_F(a_v) P_F(a_u)$ is the same across all t, M can be seen as a columnwise scaling of the Vandermonde matrix with entries $(a_v a_u)^t$. Recall that he prime product $a_v a_u$ is distinct for each edge $\{u,v\}$, so the unscaled matrix will have the Vandermonde properties, namely that any square submatrix where rows $1, \ldots, i$ are retained for some i will be invertible. Further, as each edge has that $v, u \notin F$, all columns are scaled by a nonzero value. Thus, M retains the Vandermonde properties. We now case on the size of Δ.

If $|\Delta| \le c + k - |F| + 1$, then consider submatrix of M with the first $|\Delta|$ rows, which will be invertible. If we call this submatrix M_Δ, since each element of $\boldsymbol{r}_\Delta$ is drawn from $\mathbb{F}_Q \setminus \{0\}$ we can draw

$$\boldsymbol{r}_\Delta \neq \mathbf{0} \implies M_\Delta \boldsymbol{r}_\Delta \neq \mathbf{0} \implies M \boldsymbol{r}_\Delta \neq \mathbf{0}.$$

Thus, the collision probability is 0.

If $|\Delta| > c + k - |F| + 1$, consider some $c + k - |F| + 1$ sized subset of Δ. Calling this subset Γ, we can break up the collision event as

$$M \boldsymbol{r}_\Delta = \mathbf{0} \iff M_\Gamma \boldsymbol{r}_\Gamma + M_{\Delta \setminus \Gamma} \boldsymbol{r}_{\Delta \setminus \Gamma} = \mathbf{0}$$

$$\iff \boldsymbol{r}_\Gamma = -M_\Gamma^{-1} M_{\Delta\setminus\Gamma} \boldsymbol{r}_{\Delta\setminus\Gamma}.$$

where M_Γ is the submatrix of M with columns corresponding to Γ and $\boldsymbol{r}_\Gamma$ is the subvector of M with elements corresponding to Γ. $M_{\Delta\setminus\Gamma}$ and $\boldsymbol{r}_{\Delta\setminus\Gamma}$ are defined similarly. The second biimplication comes from the observation that M_Γ is invertible.

Now, if we condition on any fixing of $\boldsymbol{r}_{\Delta\setminus\Gamma}$, the right hand side will be a fixed vector of field elements. The probability that $\boldsymbol{r}_\Gamma$ equals this fixed vector, since each element of $\boldsymbol{r}_\Gamma$ is drawn uniformly, and independent of each other and of $\boldsymbol{r}_{\Delta\setminus\Gamma}$, is

$$\left(\frac{1}{Q-1}\right)^{|\Gamma|} = \left(\frac{1}{Q-1}\right)^{c+k-|F|+1} \leq \left(\frac{1}{Q-1}\right)^{c+1} \leq n^{-c}.$$

So, in either case, the collision probability is upper bounded by n^{-c}, as desired. □

5 Upper Bounds

Now that we have set up our new sketching primitives, we demonstrate its use in solving k-vertex connectivity problem in dynamic streams and both the public and private randomness distributed sketching models, achieving the following results respectively[3].

Theorem 2. *There exists: (i) a distributed sketching protocol with public randomness and $O(k\log^3 n)$ bits of communication per node, and (ii) a one-pass turnstile streaming algorithm with $O(kn\log^3 n)$ bits of space complexity that solve the k-vertex connectivity problem with high probability.*

Theorem 3. *There exists a distributed sketching protocol with private randomness and $O(\sqrt{kn}\log^{3/2} n)$ bits of communication per node, that solves the k-vertex connectivity problem with high probability.*

5.1 Public Randomness

We first describe our protocol for Theorem 2 in the distributed sketching setting, where each node computes a linear sketch of their neighborhood and sends it to the coordinator. The coordinator then generates an output from these sketches and the shared randomness.

Our protocol is as such. Each node $v \in V$, using public randomness, computes the independent set of random graphs $\{G_{r,l}\}$ for $r \in [R]$ with a suitably large $R = \Theta(\log n)$ and $l \in \{0, \ldots, \lceil 2\log n\rceil\}$, where $G_{r,l} = (V, E_{r,l})$ and $E_{r,l}$ contains each edge in E with probability 2^{-l}. For its neighborhood in each graph $G_{r,l}$ independently, each node computes a k-sparse recovery sketch $s_{r,l}(v)$ and a fingerprint sketch $f_{r,l}(v)$ with $n^{-(k+5)}$ collision probability of its row in the signed adjacency matrix. By Theorem 1, these can be done using $O(k\log n)$ bits each. It then sends all sketches to the coordinator.

In part, the coordinator, for each set $F \subseteq V$ with $|F| \leq k$, builds a spanning forest of $G[V \setminus F]$. If any of these spanning forests are not connected, they output that G *is not* k-vertex-connected. If all are connected, they output that G *is* k-vertex connected. Consequently, the union of all spanning forests will give a certificate of k-vertex-connectivity. In contrast to the previous protocols of [GMT15; AS23], computing this does not require any subsampling. Instead, for the set of sketches sent by all nodes, the coordinator transforms them in the ways outlined in the previous section in eqs. (2) and (3), getting sketches $\tilde{s}_{F,r,l}(v)$ and $\tilde{f}_{F,r,l}(v)$ from all nodes $v \in V$.

Now, in order to build a spanning forest of $G[V \setminus F]$, the coordinator emulates a Borůvka process in R rounds. For each round $r \in [R]$, it posseses k-sparse recovery and fingerprint sketches for each blob C at

[3] Throughout this section we shall assume that $k = o(n/\log^3 n)$. Whenever $k = \Omega(n/\log^3 n)$, each vertex can simply store or send their entire neighborhood.

each level of edge subsampling l, which together form an ℓ_0 sampler sketch of the blob's boundary edges with failure probability $e^{-\Omega(k)}$ and hallucination probability $n^{-\Omega(k)}$. Thus, over R rounds, the probability of hallucination *or* that the Borůvka process fails to complete is at most $n^{-\Omega(k)}$. We show this below as Lemma 5.1. Union bounding over the at most n^{k+1} sets F of size up to k, we get that with high probability no edges are hallucinated and that every Borůvka process completes, which means that our protocol successfully constructs a spanning forest of every $G[V \setminus F]$. Specifically, with the above parameters the failure probability is n^{-2}. This can be boosted to n^{-c} for any arbitrary constant c by increasing the sketch sizes be $O((c+k)\log n)$, with no asymptotic effect. Efe: Would it be better to present big O exponents here for high probability and do the exact math in the lemma, matching private randomness? !!!

The communication complexity of the protocol is from each node sending $O(\log^2 n)$ sketches of size $O(k \log n)$ bits each, which gives a total communication of $O(k \log^3 n)$ bits per node. This achieves the result of Theorem 2.

Lemma 5.1. *For a given set F, round $r \in [R]$ in the above Borůvka process, and blob C, the sketches $\sum_{v \in C} \tilde{s}_{F,r,l}(v)$ and $\sum_{v \in C} \tilde{f}_{F,r,l}(v)$ form an ℓ_0 sampler sketch of C's boundary edges with failure probability $2e^{-k/12}$ and hallucination probability $n^{-(k+5)}$. Consequently, the probability that the Borůvka process fails to complete or produces a hallucinated edge over all R rounds is at most $n^{-(k+3)}$.*

Proof. Consider some F, r, and C. Recall that the ℓ_0 procedure goes through all subsampling levels l from $\lceil 2 \log n \rceil$ to 0, trying to recover the underlying boundary vector—call it $\boldsymbol{b}$—using $\sum_{v \in C} \tilde{s}_{F,r,l}(v)$. It then verifies that this was indeed a correctly recovered vector and not a non-sparse vector that "appeared" sparse by computing its fingerprint—call it $\overline{f}_{F,r,l}(\boldsymbol{b})$—directly as

$$\overline{f}_{F,r,l}(\boldsymbol{b})_t = \sum_{\substack{\{v,u\}:\, \boldsymbol{b}_{\{v,u\}} \neq 0 \\ v \in C, u \notin C \cup F}} \operatorname{sgn}(v-u) \boldsymbol{r}_{\{v,u\}} (a_v a_u)^t P_F(a_v) P_F(a_u),$$

and checking if $\sum_{v \in C} \tilde{f}_{F,r,l}(v) = \overline{f}_{F,r,l}(\boldsymbol{b})$.

First, we bound the probability that the underlying boundary vector is not k-sparse at any level. If $\boldsymbol{b}$ is k-sparse itself, it shall obviously be k-sparse at all levels. Otherwise, denoting $d := \operatorname{Sup}(\boldsymbol{b})$, take the level l such that $k/4d \leq 2^{-l} < k/2d$. Call the subsampled underlying vector at this level $\boldsymbol{b}_l$. Note that $\mathbb{E}[d_l := \operatorname{Sup}(\boldsymbol{b}_l)] = d2^{-l}$. We have that

$$\mathbf{Pr}[0 < d_l \leq k] = 1 - \mathbf{Pr}[d_l = 0] - \mathbf{Pr}[d_l > k].$$

Individually,

$$\mathbf{Pr}[d_l = 0] = (1-2^{-l})^d \leq e^{-d2^{-l}} \leq e^{-d\frac{k}{4d}} = e^{-k/4},$$

and

$$\mathbf{Pr}[d_l > k] \leq \mathbf{Pr}\left[d_l > 2d2^{-l}\right] = \mathbf{Pr}[d_l > 2\,\mathbb{E}[d_l]] \leq e^{-d2^{-l}/3} \leq e^{-k/12}.$$

Plugging these back in, we get

$$\mathbf{Pr}[0 < d_l \leq k] \geq 1 - e^{-k/4} - e^{-k/12} \geq 1 - 2e^{-k/12}.$$

So, the probability that $\boldsymbol{b}$ is not k-sparse at any level is at most $2e^{-k/12}$.

For the second part of the lemma, a hallucination can happen when a nonempty non-k-sparse underlying vector $\boldsymbol{b}_l$'s sparse recovery sketch collides with some actually k-sparse boundary edge vector. The probability that these two vectors also collide on the fingerprint sketch, leading to a hallucination, is $n^{-(k+5)}$. We can union bound over all $R = \Theta(\log n)$ rounds and over the at most n blobs per round to get a hallucination probability over the whole process of at most $n^{-(k+3)}$.

Now, assume that no hallucinations happen throughout the entire Borůvka process. Then, at each round a blob fails to grow with probability at most $2e^{-k/12}$, since otherwise the k-sparse recovery sketch can pull out a boundary edge.

TODO: It remains to show that with probability $1 - n^{-(k+3)}$ a spanning forest is found in $O(\log n)$ rounds. What's the neatest rigorous way of doing this? □ !!!

5.2 Private Randomness

We now explore the private randomness distributed sketching model. Here, we produce a result that might seem to be at first counterintuitive. Particularly, as our result in the public randomness model carries exactly a k-factor blowup over the regular connectivity protocol of [AGM12] and in conjunction with the existance of $\Omega(kn)$ lower bounds Efe: citation and maybe tighter lower bounds?, a reasonable assumption !!! is that k-vertex connectivity in the private randomness setting would also require a k-factor blowup over the current best regular connectivity protocol of [HKTZZ19]. This would therefore result in an $\tilde{O}(k\sqrt{n})$ bit-per-vertex protocol.

We prove this assumption wrong and achieve the result of Theorem 3 by providing a protocol with only a $\sqrt{k}$ blowup in communication cost over regular connectivity. The protocol is as such. First, for succinctness define $N := 4kn \log n$. Each node $v \in V$, using their *private* randomness, selects $\min\{\sqrt{N}, \deg(v)\}$ random neighbors. It also computes an $O(\sqrt{N})$-sparse recovery sketch $s(v)$ of its row in the signed adjacency matrix, which by Theorem 1 can be done using $O(\sqrt{N}\log n)$ bits. It then sends all random neighbors and $s(v)$ to the coordinator.

The coordinator—using the random neighbors sent by all nodes—constructs the graph H where $H = (V, E_\cup)$ and $E_\cup$ is the union of all edges arising from the random neighbors. Then, for each set $F \subseteq V$ with $|F| \leq k$, they compute the sketch $\tilde{s}_F(v)$ from each node's sent sketch $s(v)$. Now, for each component C of $H[V \setminus F]$, they take the sum $\sum_{v \in C} \tilde{s}_F(v)$ in order to get an $O(\sqrt{N})$-sparse recovery sketch of the boundary edge vector of C within $H[V \setminus F]$. Applying the result of [HKTZZ19] shows that with probability at least $1 - n^{-\Omega(k)}$, these underlying boundary edge vector s will indeed be $O(\sqrt{N})$-sparse, allowing the coordinator to decode it for each component and construct a spanning forest of $G[V \setminus F]$. We show this in detail below as Lemma 5.2. Finally, union bounding over all sets F of size up to k gives that with high probability our protocol computes a spanning forest of every $G[V \setminus F]$. The coordinator can finally look at at these and output that the graph is k-vertex-connected if all of them are connected or that the graph is not k-vertex-connected if any are disconnected. Alternatively, taking the union of all spanning forests gives a certificate of k-vertex-connectivity.

For the communication complexity of the protocol, each node sends a message containing at most $\sqrt{N}$ random neighbors, which takes $\sqrt{N}\log n$ bits to transcribe, as well as the sketch of $O(\sqrt{N}\log n/)$ bits. This puts the total communication per node at $O(\sqrt{N}\log n) = O(\sqrt{kn}\log^{3/2} n)$ bits.

Lemma 5.2. *For a given set F and component C of $H[V \setminus F]$, with probability at least $1 - n^{-k+3}$ C has $O(\sqrt{N})$ boundary edges.*

Proof. Consider some F and C, fixing $G[V \setminus F]$. Every node in this graph has sent $\min\{\sqrt{N}, \deg_G(v)\}$ of their random neighbors in G to the coordinator. At least $\min\{\sqrt{N} - k, \deg_{G[V\setminus F]}(v)\}$ of these must be in $G[V \setminus F]$. This is because if $\sqrt{N} \geq \deg_G(v)$, v sends all of its neighbors, of which $\deg_{G[V\setminus F]}(v)$ are in $G[V \setminus F]$. Alternatively, if $\sqrt{N} < \deg_G(v)$, out of the $\sqrt{N}$ neighbors it sent, at most k of them can be in F. So, $\sqrt{N} - k$ of them must be in $G[V \setminus F]$.

Now, using the fact that $N = 4kn \log n$ and $k \leq n$, $\sqrt{N} - k \geq \sqrt{N}/2$. Therefore, each node has sent at least $\min\{\sqrt{N}/2, \deg_{G[V\setminus F]}(v)\}$ random neighbors in $G[V \setminus F]$.

TODO: finish !!!

□

The tail bound result in Corollary 2.22 of [HKTZZ19] shows that with probability $1 - 2^{-\Omega(k \log n)}$, there will be at most $O(N/\sigma)$ edges of $G[V \setminus F]$ that connect two different components of $H[V \setminus F]$.

5.3 Turnstile Streams

Our public randomness algorithm above can also be implemented in the turnstile streaming model. Here, instead of each node computing a sketch from their neighborhood, the algorithm maintains each node's sketch dynamically and in parallel. As our sketches are linear, this can be done simply and efficiently on edge updates. At the end of the stream, the algorithm post-processes all sketches in the same way as the coordinator's part of the distributed protocol. Since each node's sketch is maintained simultaneously, the total storage required to emulate the distributed protocol is then $O(kn \log^3 n)$ bits. Any sketches computed in post-processing also fit into this space.

There is one additional consideration in the turnstile streaming model, which is that any randomness used int the sketching matrix—specifically for the fingerprint sketch—now needs to be stored as well. This has potential to present a problem, since each node computes and stores the random vector $\boldsymbol{r}$ of size $\binom{n}{2}$. In the streaming model, this would require $O(n^2 \log n)$ bits to store, which is too large.

There are two ways to overcome this. The first is to use Nisan's psudorandom TODO: finish this part. !!! There is—however—a second way, which highlights the generality of the annihilator sketch. The sketching matrix Φ, which is the matrix product of a Vandermonde matrix and a random diagonal matrix, can be replaced by a single fully random matrix of the same dimension. This fully random matrix then only takes $O(kn \log n)$ bits to store, which is within our budget. We now explain this second way of constructing the fingerprint sketch $f(v)$ which takes $O((c+k)\log n)$ bits of storage and has collision probability n^{-c} for any post-processed sketch $\tilde{f}_F(v)$.

An Alternate Fingerprint Sketch Construction. Each node maintains the sketch $f(v) = \Phi \boldsymbol{a}_v^{\pm}$ for a sketching matrix $\Phi \in \mathbb{F}_Q^{(c+k+1)\times n}$, where each entry $\Phi_{t,v}$ is an independently and randomly drawn field element of $\mathbb{F}_Q \setminus \{0\}$. This sketch maintains

$$f(v)_t = \sum_{u \in \mathrm{N}(v)} \mathrm{sgn}(v-u)\Phi_{t,u}$$

for each coordinate $t \in \{0, \dots, c+k\}$. We can then, given a subset $F \setminus V$ with $|F| \leq k$, take the kernel $\left\{ \boldsymbol{p} \in \mathbb{F}_Q^{c+k+1} \mid \boldsymbol{p}^\mathsf{T} \Phi_F = \boldsymbol{0} \right\}$ where Φ_F is the submatrix of Φ containing only the columns of F. This set has dimension at least $c+k+1-|F| \geq c+1$, so we can take $c+1$ linearly independent vectors $\boldsymbol{p}_0, \dots, \boldsymbol{p}_c \in \mathbb{F}_Q^{c+k+1}$ inside the kernel. For each $\boldsymbol{p}_t$, we define $z_t(v) \coloneqq \boldsymbol{p}_t^\mathsf{T} \Phi_v$ where Φ_v is the vth column of Φ. This generalizes the polynomial P_F in the Vandermonde construction, and again has that $z_j(v) = 0$ if $v \in F$. We now have the ingredients to construct the sketch $\tilde{f}_F(v)$, which is constructed as

$$\tilde{f}_F(v)_t = z_j(v) \boldsymbol{p}_t^\mathsf{T} f(v) \tag{3}$$

for each coordinate $t \in \{0, \dots, c\}$. Importantly, $\tilde{f}(v)$ can be constructed using only the kernel vectors $\boldsymbol{p}_1, \dots, \boldsymbol{p}_{c+1}$, $f(v)$, and Φ which is based on shared randomness independent of the graph. We can then make a similar observation to Lemma 4.1.

Lemma 5.3. *For any node $v \notin F$ and each coordinate $t \in \{0, \dots, c\}$,*

$$\tilde{f}_F(v)_t = \sum_{u \in N(v) \setminus F} \mathrm{sgn}(v-u) z_t(v) z_t(u).$$

Proof. We expand the definition from eq. (3) as

$$\begin{aligned} z_t(v)\boldsymbol{p}_t^\mathsf{T} f(v) &= z_t(v)\sum_{i=0}^{c+k}(\boldsymbol{p}_t)_i f(v)_i \\ &= z_t(v)\sum_{i=0}^{c+k}(\boldsymbol{p}_t)_i \sum_{u\in \mathrm{N}(v)} \operatorname{sgn}(v-u)\Phi_{i,u} \\ &= \sum_{u\in \mathrm{N}(v)} \operatorname{sgn}(v-u) z_t(v)\sum_{i=0}^{c+k}(\boldsymbol{p}_t)_i\Phi_{i,u} \\ &= \sum_{u\in \mathrm{N}(v)} \operatorname{sgn}(v-u) z_t(v) z_t(u) \\ &= \sum_{u\in \mathrm{N}(v)\setminus F} \operatorname{sgn}(v-u) z_t(v) z_t(u), \end{aligned}$$

where the last equality is because $z_t(u) = 0$ when $u \in F$. □

As with out other sketches, applying the same logic as in Lemma 4.2 cancels out the interior edges for the summed sketch of a component, giving that for any set of vertices $C \subseteq V \setminus F$ and each coordinate $t \in \{0, \dots, c\}$,

$$\sum_{v\in C} \tilde{f}_F(v)_t = \sum_{\substack{\{v,u\}\in E:\\ v\in C, u\notin C\cup F}} \operatorname{sgn}(v-u) z_t(v) z_t(u).$$

It remains to show that for a fixed F of size $|F| \le k$, the collision probability of $\sum_{v\in C} \tilde{f}_F(v)$, which is the sketch of the actual underlying boundary edge vector of C, and the sketch of a disting purported underlying boundary edge vector of C is n^{-c}. As before, call the actual boundary edge vector $\boldsymbol{b}$ and the purported boundary edge vector $\boldsymbol{b}'$ which is under an edge set E', such that $\boldsymbol{b} \neq \boldsymbol{b}'$. Call the sketch of $\boldsymbol{b}'$ $\overline{f}$.

Proof of collision probability. We shall condition on Φ_F, meaning we fix the submatrix of Φ containing only the columns of F. This fixes $\boldsymbol{p}_0, \dots, \boldsymbol{p}_c$. Now, $\boldsymbol{b} \neq \boldsymbol{b}'$ implies, as explained above, that

$$\Big\{\{v,u\} \mid \{v,u\} \in E, v \in C, u \notin C \cup F\Big\} \neq \Big\{\{v,u\} \mid \{v,u\} \in E', v \in C, u \notin C \cup F\Big\}.$$

Therefore,

$$\begin{aligned} \sum_{v\in C} \tilde{f}_F(v)_t - \overline{f}_t &= \sum_{\substack{\{v,u\}\in E\setminus E':\\ v\in C, u\notin C\cup F}} \operatorname{sgn}(v-u) z_t(v) z_t(u) - \sum_{\substack{\{v,u\}\in E'\setminus E:\\ v\in C, u\notin C\cup F}} \operatorname{sgn}(v-u) z_t(v) z_t(u) \qquad (4) \\ &= \sum_{\substack{\{v,u\}\in E\triangle E':\\ v\in C, u\notin C\cup F}} (-1)^{\mathbb{1}\left[\{v,u\}\in E'\right]} \operatorname{sgn}(v-u) z_t(v) z_t(u), \qquad (5) \end{aligned}$$

which is a polynomial of degree two in the variables $z_t(v)$ for $v \in V \setminus F$ with nonzero coefficients. Recall that $z_t(v) = \boldsymbol{p}_t^\mathsf{T}\Phi_v$ where Φ_v is the vth column of Φ. Since $\boldsymbol{p}_t$ is fixed—importantly with $\boldsymbol{p}_t \neq \boldsymbol{0}$—and Φ_v for each $v \in V \setminus F$ is a uniformly and independently random vector, each $z_t(v)$ is also uniformly and independently random. Therefore, by the Schwartz-Zippel lemma Efe: citation, $\mathbf{Pr}\left[\sum_{v\in C} \tilde{f}_F(v)_t - \overline{f}_t = 0\right] \le 2/(Q-1) \le$!!!
$O(n^{-3})$. Furthermore, since each $\boldsymbol{p}_t$ is linearly independent, $\mathbf{Pr}\left[\sum_{v\in C} \tilde{f}_F(v) - \overline{f} = \boldsymbol{0}\right] \le O(n^{-3(c+1)}) \le n^{-c}$. As this probability holds for any fixing of Φ_F, it holds unconditionally, as desired. □

6 Lower Bounds

There is one player for each vertex of an undirected graph on n labeled vertices. Each player knows its own incident edges, and the players simultaneously send deterministic messages to a referee. The referee must distinguish disconnected graphs from k-vertex-connected graphs, with no requirement on graphs satisfying neither condition. Here k-vertex-connected means that the graph has more than k vertices and remains connected after the deletion of any set of fewer than k vertices.

Theorem 4. *For every $n \geq 2$ and $1 \leq k \leq n-1$, every deterministic simultaneous-message protocol for this promise problem has worst-case maximum message length*

$$B = \Omega\left(\frac{n}{\sqrt{n}+k}\right) = \Omega\left(\min\left\{\sqrt{n}, \frac{n}{k}\right\}\right).$$

The bound holds for arbitrary encoders, without any linearity assumption.

Fix a protocol in which each message belongs to an alphabet of size at most 2^B. We may assume $B \geq 1$, since the complete graph and the empty graph require different outputs. If $B > \sqrt{n}$, the theorem already follows, so we also assume $B \leq \sqrt{n}$. All message functions below are those of this fixed n-vertex protocol.

6.1 A Hamming-distance pigeonhole lemma

Write $\mathrm{dist}(x, y)$ for Hamming distance between binary strings.

Lemma 6.1. *Let $s \geq 0$, $k \geq 1$, and $m \geq 4(s+k)$ be integers. Every map from $\{0,1\}^m$ to a set of size at most 2^s has two inputs x, y with the same image and $\mathrm{dist}(x, y) \geq k$.*

Proof. By the pigeonhole principle, at least 2^{m-s} inputs have the same output. Fix one such input x. The number of inputs at distance less than k from x, counting x itself, is

$$V = \sum_{j=0}^{k-1} \binom{m}{j} < 2^k \sum_{j=0}^{m} \binom{m}{j} 2^{-j} = 2^k (3/2)^m < 2^{k+3m/4} \leq 2^{m-s}.$$

Here we used the binomial theorem, $(3/2)^4 < 2^3$, and $m/4 \geq s + k$. Thus one of the other inputs with the same output as x has distance at least k from x. □

6.2 The graph family

The construction has two cliques and many *centers*. Each center c_i attaches to a small set inside one clique, selected by a bit a_i, and to a second small set selected by a bit b_i. When $a = b$, every center lies entirely on one side and the graph is disconnected; a center with $a_i \neq b_i$ bridges the two sides. We defeat the protocol by choosing the attachment sets so that each center's message ignores b_i, and then finding, by Lemma 6.1, two strings b that are far apart in Hamming distance yet produce identical transcripts. For general k, the attachment sets must have size at least k, so that they survive the deletion of fewer than k vertices, and the two strings must differ in at least k places, so that at least one bridging center survives.

Set

$$q = 128(\lceil \sqrt{n} \rceil + k), \qquad M = n - q.$$

(The constant is chosen so that $4k \leq q$ and $16(B + k) \leq q/8$; both inequalities are used in Section 6.5.) Suppose, toward a contradiction, that

$$M \geq 4(qB + k). \tag{6}$$

In particular, $q < n$. Reserve M vertices as *centers* $c_1, \dots, c_M$ and let U be the set of the remaining q vertices. We will partition U into four groups P_0, P_1, R_0, R_1; their sizes need not be equal. For each center c_i, we will choose subsets

$$P_i^0 \subseteq P_0, \quad P_i^1 \subseteq P_1, \quad R_i^0 \subseteq R_0, \quad R_i^1 \subseteq R_1, \qquad |P_i^0|, |P_i^1|, |R_i^0|, |R_i^1| \geq k.$$

The partition and all these subsets depend on the protocol and are fixed before the binary strings are chosen. Different centers may use overlapping subsets of the same groups.

For $a, b \in \{0,1\}^M$, define $G(a,b)$ on these vertices as follows:

- Make $P_0 \cup R_0$ a clique and $P_1 \cup R_1$ a clique.
- Give each center c_i exactly the neighborhood $P_i^{a_i} \cup R_i^{b_i}$.

There are no other edges. The two cliques together contain the q vertices of U, and the remaining $M = n - q$ vertices are the centers.

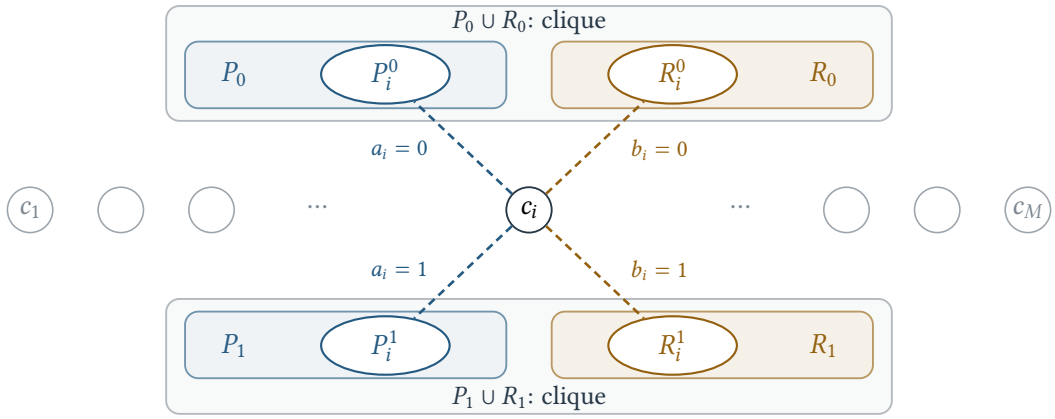


Figure 1: All centers share the same four groups; only the subsets for one center c_i are shown. The dashed connections indicate the four candidate neighbor sets of c_i: in $G(a,b)$, c_i is adjacent to every vertex of $P_i^{a_i}$ and of $R_i^{b_i}$. Each of the four subsets has at least k vertices. The other centers' attachments are omitted.

Lemma 6.2. *If $a = b$, then $G(a,b)$ has exactly two connected components. If $\mathrm{dist}(a,b) \geq k$, then $G(a,b)$ is k-vertex-connected.*

Proof. When $a = b$, each center attaches entirely to one clique: the neighborhood of c_i is $P_i^{a_i} \cup R_i^{a_i} \subseteq P_{a_i} \cup R_{a_i}$. Both cliques are nonempty, and no edge joins the two sides.

Now suppose $\mathrm{dist}(a,b) \geq k$, and delete an arbitrary vertex set X with $|X| < k$. Each clique has at least $2k$ vertices, since $P_b \cup R_b$ contains the disjoint sets P_1^b and R_1^b, each of size at least k; so each clique has a nonempty connected remainder. There are at least k centers c_i with $a_i \neq b_i$, so at least one such center survives. It has at least k neighbors in each clique, so it retains a neighbor in each remainder and joins the two remainders. Every other surviving center c_i retains a neighbor in $P_i^{a_i}$, since $|P_i^{a_i}| \geq k > |X|$. Thus $G(a,b) - X$ is connected. Since $G(a,b)$ has $n > k$ vertices, it is k-vertex-connected. □

6.3 The two local collisions and the transcript argument

For each center c_i, let $f_i(S)$ be its message when its neighborhood is $S \subseteq U$. We choose the partition and subsets so that

$$f_i(P_i^0 \cup R_i^0) = f_i(P_i^0 \cup R_i^1), \tag{7}$$

$$f_i(P_i^1 \cup R_i^0) = f_i(P_i^1 \cup R_i^1). \tag{8}$$

The common values in the two equations need not be equal. Assuming (7) and (8), the message of c_i in $G(a, b)$, namely $f_i(P_i^{a_i} \cup R_i^{b_i})$, depends on a_i but not on b_i. Sections 6.4 and 6.5 show that such a choice exists.

The other players' messages are easy to describe. A vertex in $P_0 \cup P_1$ sees its clique together with the centers c_i for which it lies in $P_i^{a_i}$, so its message depends only on a. A vertex in $R_0 \cup R_1$ sees its clique together with the centers c_i for which it lies in $R_i^{b_i}$, so its message depends only on b. There are at most q players in $R_0 \cup R_1$, so the map sending b to their joint transcript has at most 2^{qB} values. By (6), we may apply Lemma 6.1 with

$$s = qB, \qquad m = M \geq 4(s + k).$$

There are strings $x, y \in \{0,1\}^M$ with $\operatorname{dist}(x, y) \geq k$ giving the same joint transcript on $R_0 \cup R_1$ when $b = x$ and when $b = y$.

Fix $a = x$ and compare $G(x, x)$ with $G(x, y)$. The messages from $P_0 \cup P_1$ depend only on a, which is the same in both graphs. The messages from the centers also depend only on a, by (7) and (8). The joint transcript of $R_0 \cup R_1$ agrees by the choice of x and y. Thus the full transcripts agree, although $G(x, x)$ is disconnected and $G(x, y)$ is k-vertex-connected by Lemma 6.2. Both graphs satisfy the promise, so the protocol must fail on one of them.

6.4 Finding large local collision witnesses

It remains to choose the partition and the subsets so that the two collisions hold at every center simultaneously, with all four witnessing sets of size at least k. Call a partition of U into four groups P_0, P_1, R_0, R_1 (empty groups allowed) *good for* c_i if there are subsets $P_i^0 \subseteq P_0$, $P_i^1 \subseteq P_1$, $R_i^0 \subseteq R_0$, $R_i^1 \subseteq R_1$, each of size at least k, satisfying (7) and (8); otherwise call it *bad for* c_i. A partition that is good for c_i has at least k vertices in each group. In this section we show that any $16(B + k)$ vertices of U can be labeled so as to force goodness for one center; in the next we show that a random partition is good for all centers at once.

Lemma 6.3. *Set $d = 16(B + k)$. For every $Y_i \subseteq U$ of size d, there is a labeling of the vertices of Y_i by P_0, P_1, R_0, R_1 such that every partition of U extending this labeling is good for c_i.*

Proof. Since d is even, we can pair up the elements of Y_i into $d/2$ pairs. Designate k of the pairs, $\{\alpha_j, \beta_j\}$ for $j = 1, \ldots, k$, and call the remaining

$$m = d/2 - k = 8B + 7k \geq 4(2B + k)$$

pairs *free*. Consider the sets A that contain exactly one element of every pair and contain α_j rather than β_j from each designated pair. Such a set is determined by its choices on the free pairs, so these sets are indexed by $\{0,1\}^m$. For each such A, consider the ordered pair of messages

$$\big(f_i(A),\ f_i(Y_i \setminus A)\big).$$

There are at most 2^{2B} possible message pairs. Lemma 6.1 with $s = 2B$ therefore gives two sets A_i^0, A_i^1 with the same message pair whose index strings differ in at least k coordinates; that is, A_i^0 and A_i^1 make different choices on at least k free pairs. Define

$$P_i^0 = A_i^0 \cap A_i^1, \qquad P_i^1 = Y_i \setminus (A_i^0 \cup A_i^1),$$
$$R_i^0 = A_i^0 \setminus A_i^1, \qquad R_i^1 = A_i^1 \setminus A_i^0.$$

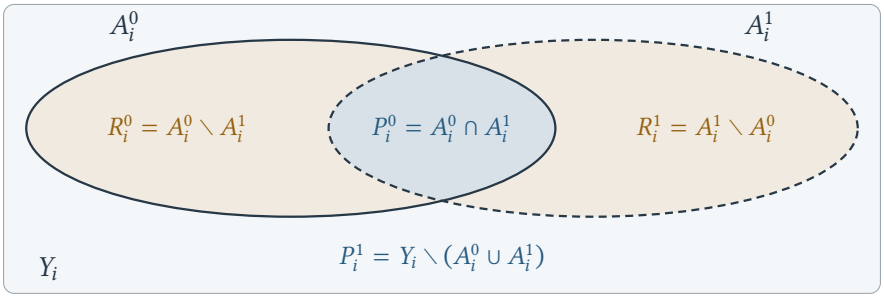


Figure 2: The two choices A_i^0, A_i^1 determine four disjoint parts of Y_i. Each designated pair places α_j in P_i^0 and β_j in P_i^1. Each free pair on which the choices differ places one vertex in R_i^0 and the other in R_i^1.

These four sets partition Y_i, as illustrated in Figure 2. Each designated pair puts α_j in P_i^0 and β_j in P_i^1, so both P sets have size at least k. Each free pair on which the two choices differ contributes one vertex to each R set, so both R sets also have size at least k.

Since $A_i^0 = P_i^0 \cup R_i^0$ and $A_i^1 = P_i^0 \cup R_i^1$, equality of the first messages gives

$$f_i(P_i^0 \cup R_i^0) = f_i(P_i^0 \cup R_i^1).$$

Since $Y_i \setminus A_i^0 = P_i^1 \cup R_i^1$ and $Y_i \setminus A_i^1 = P_i^1 \cup R_i^0$, equality of the second messages gives

$$f_i(P_i^1 \cup R_i^1) = f_i(P_i^1 \cup R_i^0).$$

Now label the vertices of P_i^0 by P_0, those of P_i^1 by P_1, those of R_i^0 by R_0, and those of R_i^1 by R_1. Every partition of U extending this labeling contains the four sets in the correct groups, so it is good for c_i. □

6.5 Counting bad partitions and concluding the bound

Assign each vertex of U independently and uniformly one of the four labels P_0, P_1, R_0, R_1. We show that with positive probability the resulting partition is good for every center. The tool is the following generalization of the Sauer–Shelah lemma to a four-letter alphabet, due to Karpovsky and Milman [KM78].

Lemma 6.4 (Karpovsky–Milman). *Let $S \subseteq \{1,2,3,4\}^q$. Suppose that for every set D of d coordinates there is a word in $\{1,2,3,4\}^D$ that is not the restriction to D of any element of S. Then*

$$|S| \le \sum_{j=0}^{d-1} \binom{q}{j} 3^{q-j}.$$

View each assignment of labels to U as a word of length q over the alphabet $\{P_0, P_1, R_0, R_1\}$, and let S be the set of words that are bad for c_i. Lemma 6.3 says exactly that on every set Y_i of d coordinates, some word is not the restriction of any element of S. Hence Lemma 6.4 bounds the number of partitions that are bad for c_i by $\sum_{j<d} \binom{q}{j} 3^{q-j}$. Dividing by the 4^q equally likely label assignments,

$$\Pr[\text{partition is bad for } c_i] \le \sum_{j=0}^{d-1} \binom{q}{j} \left(\frac{1}{4}\right)^j \left(\frac{3}{4}\right)^{q-j} = \Pr[Z < d], \qquad Z \sim \mathrm{Bin}(q, 1/4).$$

Since $B \le \sqrt{n}$, we have $d = 16(B+k) \le q/8$, so $\mu = \mathbb{E}[Z] = q/4$ satisfies $d \le \mu/2$. The Chernoff bound $\Pr[Z \le \mu/2] \le e^{-\mu/8}$ gives $\Pr[Z < d] \le e^{-q/32}$. Taking a union bound over the $M \le n$ centers,

$$\Pr[\text{partition is bad for some center}] \le n e^{-q/32} \le n e^{-4\sqrt{n}} < 1.$$

Thus some partition is good for every center. Fix it, and for each center fix the four witnessing subsets. The transcript argument of Section 6.3 then contradicts the correctness of the protocol. Every correct protocol must therefore violate (6), that is, $M < 4(qB + k)$. Since $M = n - q$, $4k \le q$, and $B \ge 1$, this gives

$$n < (4B + 1)q + 4k \le 6Bq.$$

Consequently,

$$B > \frac{n}{6q} = \Omega\left(\frac{n}{\sqrt{n} + k}\right) = \Omega\left(\min\left\{\sqrt{n}, \frac{n}{k}\right\}\right).$$

Together with the case $B > \sqrt{n}$ handled at the start, this proves Theorem 4. If messages have variable length at most B, each player has fewer than 2^{B+1} possible messages; applying the argument with $B + 1$, together with $B \ge 1$, gives the same asymptotic bound.

In particular, the square-root lower bound is preserved for $k \le \sqrt{n}$, while $k = n^{2/3}$ gives $\Omega(n^{1/3})$.

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